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Evaluating the Architectural Reasoning Capabilities of LLM Provers via the Obfuscated Natural Number Game

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1 PROBLEM

- LLM provers do well on MiniF2F and PutnamBench, but is that **genuine logical reasoning** or **semantic pattern matching** against pre-training data?
- **Architectural Reasoning**: proving theorems using only local axioms and definitions in an alien math domain, with no meaningful names, no Mathlib, and no high-level tactics.
- Discovering new mathematics needs this ability, because unexplored domains have no familiar names or libraries.

2 METHOD

SAME LEAN 4 THEOREM (`add_comm`) AS NOISE λ RISES

```
λ = 0    add_comm (a b : MyNat) : add a b = add b a
λ = 0.2  add_KmFm (a b : Mmyat) : add a b = add b a
λ = 0.4  add_Φom7 (a b : M0Nae) : add a b = add b a
λ = 0.6  jad_Qηℂm (a b : oEyab) : Eξ a b = Eξ b a
λ = 0.8  μ4dA_c2o (a b : dBNaT) : Ydd a b = Ydd b a
λ = 1.0  KΩΦzαυωT (a b : H0Ja) : KxN a b = KxN b a
```

PIPELINE

Parse NNG → Obfuscate → Query → LLM proof → Lean 4 compiler → Analyze

68 Natural Number Game problems with every definition, operator, axiom and theorem name perturbed character by character ($P = \lambda^{2.5}$). 6 noise levels, 5 models, 5 runs per theorem per level.

3 KEY RESULTS

5 / 5
models were significantly **slower** on obfuscated tasks: a universal latency tax
ANOVA on average time, $p \leq 0.0004$

3 / 5
reasoning models (DeepSeek-R1, GPT-5, DeepSeek-Prover-V2) kept their accuracy
no significant drop, $p = 0.086$ to 0.31

2 / 5
general models (GPT-4o, Claude-Sonnet-4.5) lost accuracy once names were obfuscated
significant drop, $p = 0.024$ and 0.0001

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